# Definitions and Values

Name: _____ Date: _____

As a reminder, these are some constructors provided by the book:

`Circle(int radius, String color)` `Rect(int width, int height, String color)`

**Q1.** Does each line run? If not, say what is wrong in a few words.

| definition | runs? | if not, what is wrong |
|---|---|---|
| `Pict box = new Rect(20, 30, "blue");` | | |
| `string name = "Val";` | | |
| `Pict p = Circle(25, "red");` | | |
| `int count = "5";` | | |
| `Pict dot = new Circle(25);` | | |
| `Pict p = new Circle("red", 25);` | | |

**Q2.** For each of the two definitions below, label all the parts you can with **type**, **name**, **value**, **constructor**, **argument**

```
Pict dot = new Circle(25, "red");
```

```
int year = 2026;
```

**Q3.** Write a definition for a green rectangle 10 wide and 60 tall, named `stem`.

**Q4.** Write the value each name holds after this program runs (it doesn't have errors)

```
int total = 7 * 3;
int half = total / 2;
String label = "Total: " + total;
int agePlusOne = 17 + 1;
String textPlusOne = "17" + 1;
```

`total` = _____

`half` = _____

`label` = _____

`agePlusOne` = _____

`textPlusOne` = _____

**Q5.** Two students average the scores 7 and 8. Neither line is an error.

```
int mean1 = (7 + 8) / 2;
```

```
double mean2 = (7 + 8) / 2.0;
```

`mean1` = _____ `mean2` = _____ Which belongs in a grade report? _____

**Q6.** In "Does each line run", Java would report each of the errors and refuse to run those programs. Would you have rather gotten an error from the `mean` example? Why or why not?